

Fourier series and wave packets

Ross Grogan

Derivation of the Fourier series formula

We can define an inner product on functions periodic on $[-L, L]$ by

$$\langle f, g \rangle = \int_{-L}^L f g.$$

Consider the following families of sinusoids of various frequencies:

$$c_n(x) := \cos\left(\frac{n\pi x}{L}\right), n \in \mathbb{Z}_{\geq 0} \quad \text{and} \quad s_m(x) := \sin\left(\frac{m\pi x}{L}\right), m \in \mathbb{Z}_{\geq 0}$$

By performing the necessary integrals, we can check that the s_n 's and c_m 's are orthogonal (i.e., that $\{c_n \mid n \in \mathbb{Z}_{\geq 0}\} \cup \{s_m \mid m \in \mathbb{Z}_{\geq 0}\}$ is an orthogonal set) and satisfy

$$\langle c_n, c_n \rangle = \begin{cases} L & n \neq 0 \\ 2L & n = 0 \end{cases} \quad \text{and} \quad \langle s_m, s_m \rangle = \begin{cases} L & m \neq 0 \\ 0 & m = 0 \end{cases}.$$

If we define $\hat{c}_n := c_n / \|\hat{c}_n\|$ and $\hat{s}_m := s_m / \|\hat{s}_m\|$, then the $\hat{c}_n$'s and $\hat{s}_m$'s form an orthonormal set. And if we assume this set also spans the vector space of periodic functions on $[-L, L]$, then an arbitrary periodic function f on $[-L, L]$ can be expressed as

$$f(x) = \sum_{n=0}^{\infty} \langle f, \hat{c}_n \rangle \hat{c}_n + \langle f, \hat{s}_n \rangle \hat{s}_n,$$

which, since $\|\cdot\| = \sqrt{\langle \cdot, \cdot \rangle}$, is equivalent to

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} \frac{\langle f, c_n \rangle}{\langle c_n, c_n \rangle} c_n + \frac{\langle f, s_n \rangle}{\langle s_n, s_n \rangle} s_n \\ &= \left(\frac{\langle f, c_0 \rangle}{2L} + 0 \right) + \sum_{n=1}^{\infty} \frac{\langle f, c_n \rangle}{L} c_n + \frac{\langle f, s_n \rangle}{L} s_n. \end{aligned}$$

Thus, assuming the c_n 's and s_m 's form a basis for the vector space of periodic functions on $[-L, L]$, an arbitrary periodic function f on $[-L, L]$ can be expressed as

$$f = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n c_n + b_n s_n, \quad \text{where } a_n = \frac{1}{L} \langle f, c_n \rangle \text{ and } b_n = \frac{1}{L} \langle f, s_n \rangle.$$

Fourier series with complex exponentials

In the previous section, we learned that if we assume sinusoidal functions of different frequencies form a basis for the vector space of periodic functions on $[-L, L]$, then we can express one of those periodic functions f as

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right),$$

where $a_n = \frac{1}{L}\langle f, c_n \rangle$ and $b_n = \frac{1}{L}\langle f, s_n \rangle$.

What happens when we reexpress the sinusoids by using the consequences of Euler's formula?:

$$\begin{aligned}\cos(\theta) &= \frac{\exp(i\theta) + \exp(-i\theta)}{2}, \\ \sin(\theta) &= \frac{\exp(i\theta) - \exp(-i\theta)}{2i}.\end{aligned}$$

Well, letting $\theta_n = n\pi x/L$ for convenience, we have

$$\begin{aligned}a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) &= a_n \cos(\theta_n) + b_n \sin(\theta_n) \\ &= \frac{1}{2}\left(a_n \frac{\exp(i\theta_n) + \exp(-i\theta_n)}{2} + b_n \frac{\exp(i\theta_n) - \exp(-i\theta_n)}{2i}\right) \\ &= \frac{1}{2}a_n(\exp(i\theta_n) + \exp(-i\theta_n)) - \frac{1}{2}ib_n(\exp(i\theta_n) - \exp(-i\theta_n)) \\ &= \frac{1}{2}(a_n + ib_n)\exp(-i\theta_n) + \frac{1}{2}(a_n - ib_n)\exp(i\theta_n).\end{aligned}$$

This leads to f being reexpressed as

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} \frac{1}{2}(a_n + ib_n)\exp(-i\theta_n) + \sum_{n=1}^{\infty} \frac{1}{2}(a_n - ib_n)\exp(i\theta_n).$$

The idea now is to associate positive indices with the positive complex frequencies $i\theta_n$ and negative indices with the negative complex frequencies. To do so, define

$$z_n := \begin{cases} \frac{1}{2}(a_n - ib_n) & n \geq 1 \\ \frac{1}{2}(a_{|n|} + ib_{|n|}) & n \leq -1 \end{cases}.$$

Now, we change the indices in the first sum. Using the fact that $-i\theta_{|n|} = -i\theta_{-n} = i\theta_n$ when $n \leq -1$, we have

$$f(x) = \frac{1}{2}a_0 + \sum_{n=-\infty}^{-1} z_n \exp(i\theta_n) + \sum_{n=1}^{\infty} z_n \exp(i\theta_n).$$

We can now see a major advantage of the negative indices: the change of indices in the first sum resulted in its argument changing to match that of the second summation.

Now, if we additionally had $\frac{1}{2}a_0 = z_0 \exp(i\theta_0)$, then the above would collapse into one compact formula:

$$f(x) = \sum_{n=-\infty}^{\infty} z_n \exp(i\theta_n).$$

Let's check this. When we use the definitions of a_n and b_n to compute z_n in the case $n \geq 1$, when $z_n = \frac{1}{2}(a_n - ib_n)$, we find

$$\begin{aligned}
z_n &= \frac{1}{2} \left(\frac{1}{L} \langle f, c_n \rangle - i \frac{1}{L} \langle f, s_n \rangle \right) \\
&= \frac{1}{2L} \langle f, c_n - i s_n \rangle \\
&= \frac{1}{2L} \langle f, \exp(-i\theta_n) \rangle.
\end{aligned}$$

In the case $n \leq -1$, when $z_n = \frac{1}{2}(a_n + ib_n)$, we actually obtain the same thing as a result of using negative indices:

$$\begin{aligned}
z_n &= \frac{1}{2} \left(\frac{1}{L} \langle f, c_{|n|} \rangle + i \frac{1}{L} \langle f, s_{|n|} \rangle \right) \\
&= \frac{1}{2L} \langle f, c_{|n|} + i s_{|n|} \rangle \\
&= \frac{1}{2L} \langle f, \exp(i\theta_{|n|}) \rangle \\
&= \frac{1}{2L} \langle f, \exp(-i\theta_n) \rangle.
\end{aligned}$$

Thus, we have

$$z_n = \frac{1}{2L} \langle f, \exp(-i\theta_n) \rangle = \frac{1}{2L} \left\langle f, \exp \left(-i \frac{n\pi x}{L} \right) \right\rangle.$$

As we hoped, we do have $z_0 \exp(i\theta_0) = \frac{1}{2}a_0$:

$$z_0 \exp(i\theta_0) = z_0 \exp(0) = z_0 = \frac{1}{2L} \langle f, \exp(0) \rangle = \frac{1}{2L} \langle f, 1 \rangle = \frac{1}{2L} \langle f, \cos \left(\frac{0\pi x}{2L} \right) \rangle = \frac{1}{2L} \langle f, c_0 \rangle = \frac{1}{2}a_0.$$

So we have the nice formula

$$\begin{aligned}
f(x) &= \sum_{n=-\infty}^{\infty} z_n \exp \left(i \frac{n\pi x}{L} \right), \text{ where} \\
z_n &= \frac{1}{2L} \left\langle f, \exp \left(-i \frac{n\pi x}{L} \right) \right\rangle.
\end{aligned}$$

The Fourier transform

Apply Fourier series to non-periodic functions by letting the period L go to ∞

Wave packets

Theorem 1. Consider a position wavefunction $\psi : \mathbb{R} \rightarrow \mathbb{C}$, and let $\phi = \mathcal{F}(\psi)$ be its Fourier transform. By Parseval's theorem, $|\phi|^2$ is also a probability density.

If we let $\langle x \rangle$ and $(\Delta x)^2$ denote the mean and variance of the probability density $|\psi|^2$, and let $\langle k \rangle$ and $(\Delta k)^2$ denote the mean and variance of the probability density $|\phi|^2$, then, assuming the appropriate integrals are defined, we have

$$\Delta x \Delta k \geq \frac{1}{2},$$

with equality if and only if ψ is an unchirped Gaussian with mean $\langle x \rangle$.

TO-DO.

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